# ZOHO INVENTORY & ERP SOLUTIONS SQUAD

ENTERPRISE CATALOG

Catalog Taxonomy • Composite Assemblies • Supplier Costs • Stock Tracking • Zoho Books Sync

**Client:** CMJ Door & Hardware Corp.
**Industry:** Commercial Doors, Architectural Hardware & Building Materials
**Model:** 3-Specialist Solutions Squad
**Commercials:** Transparent Fixed-Scope Milestones (Post-Discovery)

## Comprehensive Zoho Inventory Implementation & SKU Architecture

A dedicated solutions squad providing end-to-end catalog restructuring, intelligent SKU taxonomy, automated supplier price & markup modeling, multi-component composite door/hardware packages (BOM), duplicate resolution, real-time stock tracking, and seamless financial synchronization with Zoho Books for CMJ Door & Hardware Corp.

Focus: **Doors, Hardware & Building Materials** | Taxonomy: **Structured SKUs & Variant Attributes**

Assemblies: **Composite Items (BOM) & Package Kits** | Pricing: **Supplier Cost Tracking & Dynamic Markups**

Accounting: **Automated Zoho Books Sync & COGS**

## 1. EXECUTIVE SUMMARY & DELIVERY APPROACH

Engineered Specifically for Building Materials & Hardware Distribution

### The Door & Hardware Inventory Challenge

DOMAIN SPECIFICS

Commercial door and architectural hardware catalogs possess distinct complexities that generic inventory setups fail to handle:

- **Complex Matrix Variations:** Handing (LH/RH/LHR/RHR), fire ratings (20/45/60/90-min), core types (honeycomb, solid wood, mineral), dimensions, and finishes (US26D, US10B, Black).
- **Package Assemblies (Kits):** Pre-hung door packages consisting of a slab, frame, hinges, lockset, closer, kickplate, and threshold sold as a unified unit but tracked as separate inventory items.
- **Fluctuating Supplier Pricing:** Volatile steel/lumber costs and complex vendor price lists requiring dynamic markup formulas to preserve contractor and retail gross margins.

### Our Solution Architecture for CMJ

EXECUTION SQUAD

Rather than a lone consultant attempting data cleanup while building workflows, our **3-specialist engineering squad** handles data cleansing, catalog modeling, and accounting integration concurrently:

- **Clean Canonical Catalog:** Programmatic audit and deduplication of existing items into standardized, readable SKU conventions.
- **Composite Item & BOM Workflows:** Automated bill-of-materials deduction so assembling or selling a door combo instantly updates component stock levels.
- **Bi-Directional Financial Integrity:** Seamless Zoho Books sync ensuring item sales, landed freight costs, and COGS reflect accurately without double-entry.

## 2. SCOPE OF WORK ALIGNMENT MATRIX

Direct Mapping to CMJ Door & Hardware Corp. Requirements

| CMJ REQUESTED REQUIREMENT | SQUAD IMPLEMENTATION & ARCHITECTURE STRATEGY | KEY DELIVERABLES FOR CMJ |
|---|---|---|
| **Product & Variation Organization** | Group products into logical Item Groups (Wood Doors, Hollow Metal, Hinges, Locks, Closers) with structured attributes (Size, Finish, Handing, Rating). | Clean matrix item structure eliminating standalone clutter; standardized naming conventions across all items. |
| **Intelligent SKU Architecture** | Design a hierarchical, human-readable SKU schema incorporating Category, Subtype, Material, Dimension, and Finish codes. | Universal SKU Master Registry with clear rules for future staff to generate consistent SKUs easily. |
| **Supplier Costs & Markup Modeling** | Configure vendor price lists, landed cost allocations (freight/tariffs), and automated markup percentage rules per customer tier (Trade, Retail, Contractor). | Real-time profit margin visibility, price lists assigned to customer types, and margin floor protection. |
| **Composite Items & Product Combos** | Build Composite Items (BOM) for complete door units and hardware sets. Configure auto-assembly and component inventory deductions. | 1-click quoting and sales of pre-assembled door packages while maintaining accurate component counts. |
| **Duplicate Resolution & Cleanup** | Run automated data matching scripts to identify, merge, or archive duplicate records while preserving historical transactions. | Consolidated, deduplicated database with zero transaction orphan records or broken history. |
| **Stock Tracking & Fulfillment** | Set up multi-location stock tracking, reorder points, low-stock warnings, purchase orders, dropship routing, and pick/pack/ship workflows. | Automated purchase alerts, purchase order creation, package slips, and shipping label integrations. |
| **Zoho Books Sync & Automation** | Map chart of accounts (Inventory Asset, COGS, Sales Income), automate tax codes, and deploy custom Deluge scripts for margin alerts and reorders. | Automated real-time financial synchronization with Zoho Books; automated email/Slack alerts for low-margin quotes. |

# MEET THE INVENTORY SQUAD SQUAD ROLES

Complementary Engineering, Data Architecture & Project Governance Dedicated to CMJ

**Engagement Model:** Coordinated 3-Person Team
**Lead Contact:** Viranga Lakshan
**Timezone:** UTC+5:30 (Extensive Global Overlap)

## 3. SQUAD ENGINEERING & MANAGEMENT PROFILES

Solutions Architect + Inventory Database Engineer + Project Manager

### Viranga Lakshan
**LEAD SOLUTIONS ARCHITECT**

Associate Software Engineer, Arimac Lanka • 4+ Years Enterprise Experience

Leads systems integration, Zoho ecosystem architecture, business workflow automations, and Zoho Books financial reconciliation. Expert in ERP data pipelines, REST API workflows, and secure multi-tier catalog modeling.

- Architected product, order, tax, shipping, and tenant inventory services for enterprise digital platforms.
- Integrated mission-critical microservices with enterprise ERP (SAP) and AWS cloud infrastructure.
- Engineered automated validation workflows and custom integration middleware with error-recovery mechanisms.

Zoho Inventory | Zoho Books Sync | Deluge Scripting | ERP Integration | Workflow Automation | REST APIs

### Janith Kavishka
**INVENTORY DATABASE ENGINEER**

Programmer, DMS Electronics • Enterprise Stock Tracking & Catalog Systems

Specializes in inventory data modeling, bulk data transformations, programmatic deduplication scripts, and multi-tier distribution tracking. Experienced in managing complex product matrices and automated reconciliation.

- Contributed to an enterprise inventory and distribution tracking platform integrated with AWS and Amazon RDS.
- Architected *InvoiceIQ* accounts payable matching engine with deterministic 3-way line item reconciliation.
- Engineered *Nexware* warehouse management system with supplier purchase orders and stock flow controls.

SKU Architecture | Composite Items / BOM | Catalog Deduplication | Batch Tracking | Data Transformation | SQL / ETL

### Lithumitha Ravikumar
**PROJECT MANAGER**

Associate Project Manager, Arimac Lanka • Enterprise SaaS & Data Platforms

Manages catalog data governance, requirement specifications, sprint planning, client communication, and user training. Ensures strict milestone delivery, quality assurance, and comprehensive documentation for CMJ staff.

- Managed delivery across 4 engineering squads for enterprise multi-tenant digital commerce SaaS platforms.
- Led delivery of Castle Visualizer enterprise data platform consolidating disparate operational systems.
- Authored comprehensive standard operating procedures (SOPs), user training guides, and acceptance criteria.

Project Management | Catalog Governance | Sprint Coordination | User Training / SOPs | QA Acceptance | Milestone Tracking

## 4. DOMAIN BLUEPRINT: DOORS, HARDWARE & COMPOSITE PACKAGES

Tailored Specifically to Building Materials Distribution

### Hierarchical SKU Taxonomy Formula STANDARDIZATION

We replace ambiguous item descriptions with an intelligent, standardized SKU format that informs staff immediately of the product's exact specifications:

```
[CAT]-[TYPE]-[MAT]-[DIM]-[HAND]-[FINISH]
• Door Slab: DR-WD-SLD-3070-RH-RAW (Wood, Solid Core, 3'0"x7'0", Right Hand)
• Steel Frame: FR-HM-KD-3070-534-GRY (Hollow Metal, Knock-Down, 5-3/4" Jamb)
• Lockset: HW-LCK-MRT-ENT-US26D (Hardware, Mortise Entry, Satin Chrome)
• Commercial Closer: HW-CLS-HD-DA-ALM (Hardware, Heavy Duty Closer, Aluminum)
```

Enables instant barcode scanning, rapid warehouse bin picking, and zero confusion between field installers and office sales staff.

### Composite Items & Automated Bill of Materials (BOM) ASSEMBLIES

In Zoho Inventory, we configure **Composite Items** for pre-hung door units, commercial openings, and packaged hardware sets:

```
COMMERCIAL OPENING PACKAGE (SKU: PKG-COMM-3070-01)
├── 1x Door Slab (DR-WD-SLD-3070-RH-RAW)
├── 1x Hollow Metal Frame (FR-HM-KD-3070-534-GRY)
├── 3x Ball Bearing Hinges 4.5"x4.5" (HW-HNG-BB-4545-US26D)
├── 1x Heavy Duty Lever Lockset (HW-LCK-CYL-US26D)
└── 1x Heavy Duty Surface Closer (HW-CLS-HD-DA-ALM)
```

**Auto-Assembly / Auto-Disassembly:** Selling 1 package automatically decrements 1 slab, 1 frame, 3 hinges, 1 lock, and 1 closer in real-time, preventing stockout surprises.

# DEDUPLICATION, STOCKING & ACCOUNTING

Eliminating Inventory Clutter, Landed Costs, and Real-Time Zoho Books Synchronization

INTEGRITY

**Data Cleansing:** Automated & Manual Verification
**Accounting Sync:** Bi-Directional Chart of Accounts
**Custom Logic:** Deluge Automations & Webhooks

## 5. CATALOG CLEANSING, PRICING RULES & ZOHO BOOKS INTEGRATION

Building a Rock-Solid Operational Foundation for CMJ

### Programmatic Deduplication & Cleanup

ZERO DATA LOSS

Duplicates happen when different employees enter items slightly differently (e.g., "*3/0 x 7/0 Solid Core*" vs "*3070 SC Wood Door*"). Our approach cleans this without corrupting past records:

- **Automated Data Audit:** Export existing inventory, run similarity-matching algorithms on descriptions, vendor parts, and attributes to identify all duplicate clusters.
- **Canonical SKU Mapping:** Designate the true master SKU for each product group; consolidate stock counts and supplier historical costs.
- **Safe Inactivation / Merging:** Merge active records where appropriate or inactivate legacy duplicates while retaining full audit history for accounting and past invoices.

### Supplier Costs, Landed Costs & Markups

MARGIN PROTECTION

Ensure every quote and sale protects CMJ's profit margins through dynamic pricing structures:

- **Supplier Price Lists:** Manage multi-vendor cost sheets. Easily update vendor price hikes via CSV imports without breaking sales prices.
- **Landed Cost Tracking:** Distribute incoming freight, crating, and delivery charges across purchased items, calculating true cost per unit.
- **Tiered Price Rules:** Configure automated markup rules in Zoho Inventory for General Contractors, Subcontractors, Architects, and Over-the-Counter Retail.

### Warehouse Stock Tracking & Purchasing

END-TO-END FLOW

Streamline CMJ's daily physical and administrative operations:

- **Reorder Points & Safety Stock:** Establish automated minimum stock levels based on vendor lead times. Generate Purchase Orders with 1 click.
- **Multi-Location / Bin Tracking:** Track items across main warehouse, staging areas, and job-site drop-ships.
- **Fulfillment Workflows:** Seamless transition from Sales Order → Package Slip → Shipping Notice → Customer Delivery Receipt.

### Bi-Directional Zoho Books Synchronization

FINANCIAL AUDIT

Ensure complete harmony between warehouse operations and company financials:

- **Chart of Accounts Mapping:** Map Inventory Asset accounts, Cost of Goods Sold (COGS), Purchase Discrepancies, and Sales Income correctly.
- **Real-Time COGS Posting:** When an order is fulfilled in Zoho Inventory, Zoho Books immediately reflects the accurate COGS entry.
- **Vendor Bills & Payments:** Receiving items in the warehouse automatically generates or updates Bills in Zoho Books for accounts payable approval.

## 6. CUSTOM DELUGE AUTOMATIONS & MANAGEMENT REPORTS

Eliminating Manual Repetitive Tasks

### Low-Margin Guardrail

Custom Deluge script that flags any estimate or sales order where gross profit drops below CMJ's target threshold (e.g., < 25%), requiring manager approval before issuing.

### Supplier Price Hike Alert

Automated notification triggered when vendor purchasing costs increase, prompting instant review of affected sales prices and composite package rates.

### Hardware Dead-Stock Report

Configured analytics dashboard highlighting non-moving hinges, locksets, and specialized trim, empowering smart discounting and inventory liquidity.

# IMPLEMENTATION ROADMAP & PRECEDENTS

Structured Phasing, Verified Enterprise Experience, and Next Steps for CMJ

EXECUTION

**Execution Model:** 5-Phase Structured Rollout
**Handover:** Full Documentation & Team Training
**Support:** 30 Days Post-Deployment Warranty

## 7. PHASED IMPLEMENTATION ROADMAP

Step-by-Step Delivery Ensuring Zero Business Interruption

### Phase 1: Catalog Audit & Clean Data Architecture

WEEKS 1–2

- Ingest CMJ's existing product spreadsheets, vendor price books, and current Zoho setup.
- Run programmatic duplicate detection; identify and map duplicate item clusters.
- Finalize the standardized SKU naming taxonomy and Item Group matrix for doors and hardware.
- Designate clean master import templates for products, variants, and custom fields.

### Phase 2: Composite Bundles & Supplier Costing

WEEKS 2–3

- Build Composite Item packages (Pre-hung doors, complete commercial lockset trims, frame kits).
- Upload vendor cost lists; configure landed cost allocations for freight and delivery.
- Establish customer tier pricing rules (General Contractors, Retail, Commercial Subcontractors).
- Test auto-assembly and auto-disassembly component deduction workflows in a sandbox.

### Phase 3: Stocking, Fulfillment & Zoho Books Sync

WEEKS 3–4

- Configure warehouse locations, bin labeling logic, reorder points, and automated PO generation.
- Integrate and test bi-directional Zoho Books synchronization: Chart of Accounts, Asset accounts, COGS, and Tax rules.
- Establish sales order fulfillment workflows: picking, packing, shipping, and drop-ship routing.
- Conduct end-to-end transaction test from Quote → Sales Order → Fulfillment → Invoice → Payment.

### Phase 4: Deluge Automations, Reports & Training

WEEKS 4–5

- Deploy custom Deluge scripts for margin floor protection and automated reorder alerts.
- Build customized management reports: inventory valuation, sales by door/hardware category, and vendor lead times.
- Deliver live, recorded training sessions for CMJ sales, warehouse, and accounting personnel.
- Provide step-by-step Standard Operating Procedure (SOP) manuals for creating future items and bundles.

## 8. VERIFIED ENTERPRISE PRECEDENTS

Proven Track Record in Complex Inventory, ERP & Catalog Systems

### SINGER Sri Lanka PLC (DMS Electronics)

ENTERPRISE STOCK

**Multi-Tier Inventory & Distribution Tracking:** Built distribution tracking platform integrated with AWS Lambda and Amazon RDS supporting multi-tier stock flows across a nationwide distribution network, handling batch tracking and automated replenishment.

### DIMO Lanka Enterprise Gate Pass System

ERP INTEGRATION

**SAP ERP & Warehouse Logistics Integration:** Engineered vehicle and equipment dispatch platform integrated with enterprise SAP ERP, multi-role approval workflows, and audit logging for premier industrial conglomerate.

## 9. ENGAGEMENT TERMS & NEXT STEPS

Getting Started with CMJ Door & Hardware Corp.

### 1. Catalog Discovery

A 20-minute call to review your current Zoho setup, approximate SKU volume, and sample door/hardware price sheets.

### 2. Fixed Quote & Roadmap

Within 24 hours of discovery, receive an exact milestone breakdown with defined deliverables and transparent pricing.

### 3. Immediate Kickoff

Squad begins data audit and SKU design immediately upon kickoff, delivering clean catalog structure in Days 1–10.

**Ready to Organize CMJ's Inventory**

We are available to start immediately. Please let us know a convenient time for a discovery discussion, or share a sample catalog export so we can review your data structure upfront.